Parks Meteo Optimization

PEINP Meteorological Data Pipeline - Final Report

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github.com/Pegjoey99/parks-meteo-optimization

Project Overview

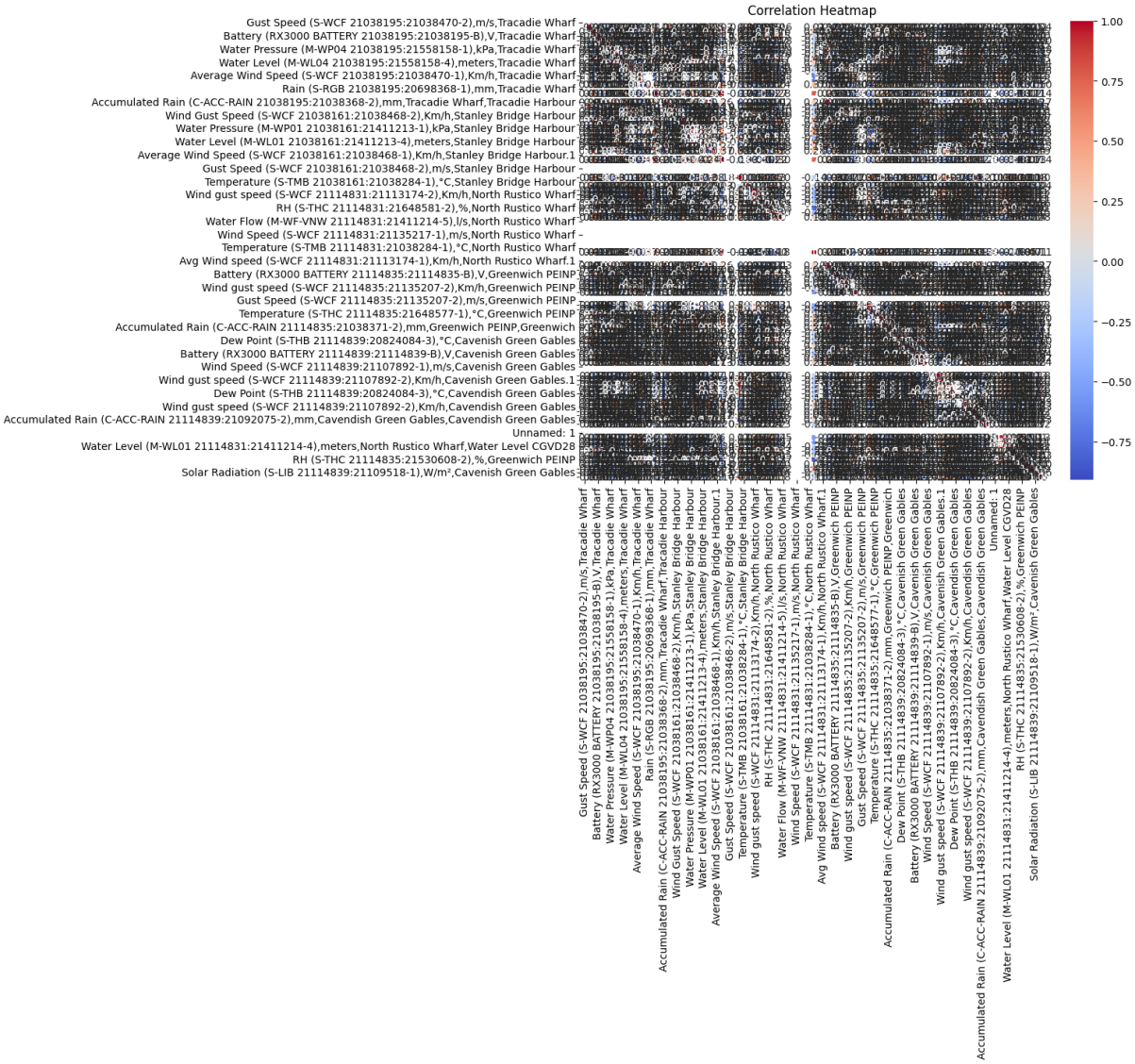
- End-to-end pipeline for meteorological data analysis in Prince Edward Island National Park
- 5 HOBO weather stations: Cavendish, Greenwich, North Rustico Wharf, Stanley Bridge Wharf, Tracadie Wharf
- ECCC Stanhope reference station (Climate ID 8300590) for cross-validation
- Agent-based architecture: Ingest > Clean > Redundancy > FWI > Uncertainty

Data Ingestion & Cleaning

- Ingested 5,086 daily observations across 20 variables
- 100% row retention after cleaning (5,086 > 5,086 rows)
- Column reduction: 20 to 16 columns (dropped redundant/empty fields)
- 2 temperature range issues flagged and handled
- Output: cleaned_data.parquet (380 KB)

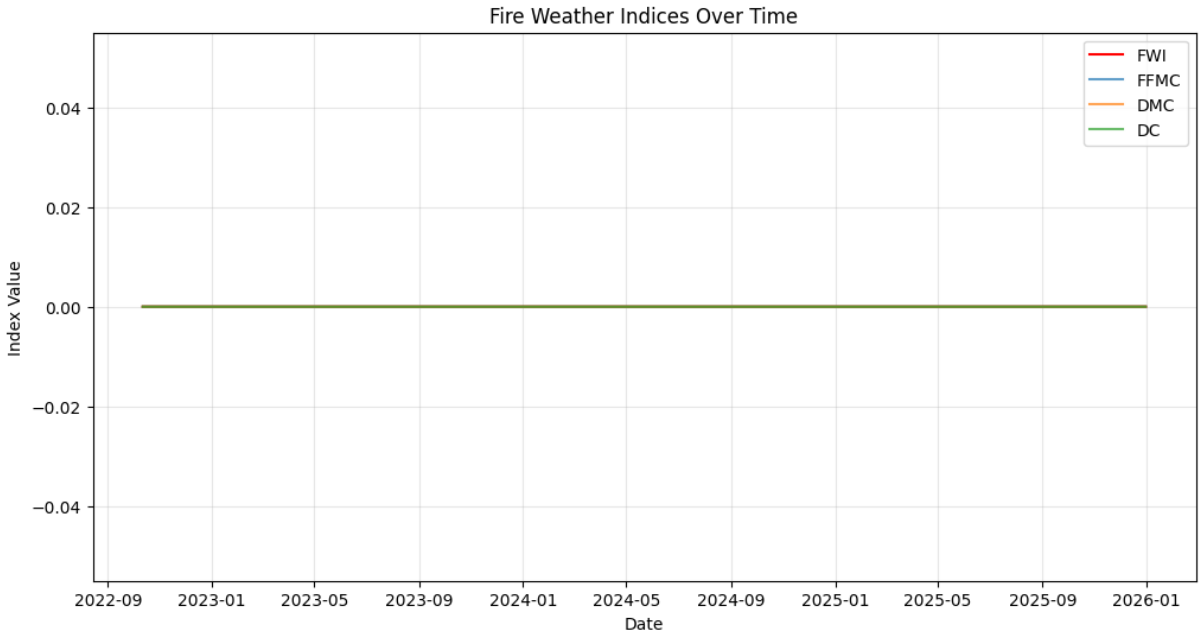
Redundancy Analysis (PCA & Clustering)

- PCA explained variance: PC1=23.1%, PC2=15.3%, PC3=14.6%
- ~7 components needed for >=90% cumulative variance
- K-Means clustering: 3 clusters (351, 2,322, 2,413 samples)
- Highest correlation: wind_speed / gust_speed (r = 0.96)
- Recommendation: review low-contribution sensors for potential decommissioning



Fire Weather Index (FWI)

- Van Wagner (1987) Canadian FWI System - 6 indices computed daily
- FWI range: 0.68 - 55.79 (mean 13.06)
- All 6 indices passed physical range validation (PASSED)
- Peak FWI = 55.8 at Stanley Bridge Wharf
- High FWI days (>=10): significant count; Very-High (>=20): flagged for review



FWI Cross-Validation vs Stanhope

- Independent validation against ECCC Stanhope reference (368 fire-season days)
- FFMC MAE = 0.025 - near-identical
- DMC MAE = 0.70, DC MAE = 53.29 (cumulative index, expected divergence)
- ISI MAE = 0.015, BUI MAE = 3.20
- FWI MAE = 0.37 - excellent agreement

Uncertainty Modeling (KDE)

- Total Variation (TV) distance: measures cleaning impact on distributions
- Highest impact: water_level (TV=0.479), water_temperature (TV=0.420)
- Moderate impact: barometric_pressure, diff_pressure, humidity, dew_point
- Lowest impact: rain (TV=0.011) - sensor producing consistently clean data
- Recommendation: investigate calibration for high-TV sensors

Key Recommendations

- 1. PCA shows 7 components capture >=90% variance from 15 variables - review low-contribution sensors
- 2. Very-High FWI days (>=20) detected - cross-reference with fire preparedness records
- 3. High cleaning impact (TV > 0.10) on water_level, water_temp - check sensor calibration
- 4. Rain sensor (TV = 0.011) producing high-quality data consistently
- 5. Consider reduced sampling frequency for redundant channels to lower maintenance costs

Repository & Reproducibility

- GitHub: github.com/Pegjoey99/parks-meteo-optimization
- Jupyter notebook: notebooks/analysis.ipynb - full end-to-end pipeline
- All outputs reproducible via: Run All in VS Code / Jupyter
- Dependencies: pip install -r requirements.txt
- License: MIT